

AI Requirement Analyzer - Requirement Analysis Report

1. Detected Ambiguities

Issue 1

Statement: It should rank them based on relevance to the job description.

Problem: The term 'relevance' is subjective and lacks a specific algorithm, scoring metric, or weighting criteria for how alignment with a job description is calculated.

Clarification Question: What specific criteria, keyword weights, or semantic scoring metrics should be used to calculate a candidate's relevance score against a job description?

Issue 2

Statement: Mainly skills and experience. And overall profile strength.

Problem: 'Overall profile strength' is not defined, and the relative importance (weighting) between skills versus experience is unspecified.

Clarification Question: How should 'overall profile strength' be quantified, and what are the exact percentage weights for skills versus years of experience?

Issue 3

Statement: Things like good companies, solid projects, impactful work.

Problem: Qualifiers like 'good', 'solid', and 'impactful' are open to interpretation and cannot be directly parsed without explicit indicators or whitelist/tier lists.

Clarification Question: What deterministic parameters define a 'good' company, a 'solid' project, or 'impactful' work (e.g., specific company tiers, project metrics, or keywords)?

Issue 4

Statement: Yes, but not strictly. Sometimes a strong fresher is better than someone with 5 average years.

Problem: It introduces conflicting business logic between strict experience duration and qualitative candidate strength without a formal rule for trade-offs.

Clarification Question: What quantifiable thresholds or scoring logic should allow a candidate with lower experience to outrank a candidate with more years of experience?

Issue 5

Statement: It should be good enough so that HR trusts it.

Problem: 'Good enough' and 'HR trust' are qualitative acceptance criteria rather than measurable machine learning performance metrics.

Clarification Question: What target values for precision, recall, top-K accuracy, or false positive/negative rates define acceptable performance for deployment?

Issue 6

Statement: Yes, that would be useful.

Problem: It is unclear whether explainability is a mandatory non-functional requirement or an optional feature, as well as what depth of explanation is required.

Clarification Question: Is explainability a strict MVP requirement, and what specific format (e.g., key match highlights, SHAP/LIME feature importance scores) should be displayed to HR?

Issue 7

Statement: Yes, we must avoid bias, especially related to gender or college background.

Problem: It specifies a high-level goal without defining how bias mitigation will be measured, constrained, or audited.

Clarification Question: Which fairness metrics (e.g., demographic parity, equalized odds) should be enforced, and what data fields must be masked during evaluation?

Issue 8

Statement: It shouldn't be slow.

Problem: 'Slow' is relative and provides no functional performance requirement for throughput or latency.

Clarification Question: What is the expected processing architecture (real-time vs. batch) and maximum acceptable processing latency per batch or single resume?

Issue 9

Statement: Ideally quick.

Problem: 'Ideally quick' lacks an explicit SLA or non-functional latency metric.

Clarification Question: What is the maximum allowed response time in seconds or milliseconds per resume under peak load?

Issue 10

Statement: We need an MVP soon

Problem: 'Soon' is an undefined timeline, making project scheduling, scope management, and milestone tracking impossible.

Clarification Question: What is the specific target date or calendar deadline for the MVP delivery?

2. Requirements Without Clarification

```
{
  "functional_requirements": [
    {
      "ambiguity_level": "Low",
      "category": "Candidate Ranking",
      "id": "FR-001",
      "measurable_criteria": "Given a job description and a set of candidate resumes, the system outputs an ordered list of shortlisted candidates within 100% of execution test runs.",
      "requirement": "The system shall automatically shortlist and rank software engineering candidates based on their relevance to a provided job description.",
      "source": "Hiring Manager: 'automatically shortlist candidates... rank them based on relevance to the job description'"
    },
    {
      "ambiguity_level": "Medium",
      "category": "Candidate Evaluation",
      "id": "FR-002",
      "measurable_criteria": "The feature extraction module parses and weights candidate skills, experience length, company tier, and project impact metrics during scoring.",
      "requirement": "The system shall evaluate candidate relevance based on extracted skills, years of experience, and profile attributes such as company reputation and project impact.",
      "source": "Hiring Manager: 'Mainly skills and experience. And overall profile strength... good companies, solid projects, impactful work'"
    },
    {
      "ambiguity_level": "Medium",
      "category": "Scoring Logic",
      "id": "FR-003",
```

```

"measurable_criteria": "In automated test cases, a high-performing fresher profile with top skills/projects scores higher than a 5-year profile with average skills/projects.",
"requirement": "The candidate scoring model shall balance years of experience against skill level and project impact so that highly capable entry-level candidates can outperform candidates with longer average experience.",
"source": "Hiring Manager: 'Yes, but not strictly. Sometimes a strong fresher is better than someone with 5 average years.'"
},
{
"ambiguity_level": "Medium",
"category": "Explainability",
"id": "FR-004",
"measurable_criteria": "Each ranked candidate output includes associated scoring factors (e.g., top key matching skills, experience score, project impact weight).",
"requirement": "The system shall provide rationale or explanations detailing why a specific candidate was assigned their rank.",
"source": "ML Engineer & Hiring Manager: 'Do we need explainability? ... Yes, that would be useful.'"
},
{
"ambiguity_level": "Medium",
"category": "Data Ingestion",
"id": "FR-005",
"measurable_criteria": "Data ingestion pipeline successfully processes unstructured historical resume files and converts historical hiring decisions into training feature sets.",
"requirement": "The system shall ingest and parse unstructured historical resumes and prior hiring decision data for model training and evaluation.",
"source": "Hiring Manager & ML Engineer: 'Do we have historical hiring data... past resumes and hiring decisions, but they\u2019re not very structured.'"
}
],
"non_functional_requirements": [
{
"ambiguity_level": "Medium",
"category": "Fairness",
"id": "NFR-001",
"measurable_criteria": "Demographic parity and adverse impact ratio across gender and university tiers must satisfy the 80% rule (disparate impact ratio >= 0.80) during bias auditing.",
"requirement": "The system must prevent bias in candidate ranking, specifically ensuring non-discrimination based on gender and college background.",
"source": "Hiring Manager: 'avoid bias, especially related to gender or college background.'"
},
{
"ambiguity_level": "High",
"category": "Accuracy",
"id": "NFR-002",
"measurable_criteria": "Ranking alignment rate against human HR recruiter selections reaches an agreed precision threshold (e.g., Top-K overlap >= 80% on test dataset).",
"requirement": "The system shall achieve candidate ranking accuracy sufficient to earn the trust of the HR team.",
"source": "Hiring Manager: 'It should be good enough so that HR trusts it.'"
},
{
"ambiguity_level": "High",
"category": "Performance",
"id": "NFR-003",
"measurable_criteria": "Average processing and ranking time per resume is under a target SLA (e.g., < 2 seconds per resume).",
"requirement": "The system shall process and evaluate resumes quickly to ensure low response times.",
"source": "Hiring Manager: 'It shouldn\u2019t be slow... Ideally quick.'"
},
{
"ambiguity_level": "Low",
"category": "Explainability",
"id": "NFR-004",
"measurable_criteria": "Model outputs include SHAP/LIME values or feature importance scores for 100% of ranked candidates.",
"requirement": "The machine learning scoring model must yield interpretable decision outputs for transparency.",
"source": "ML Engineer & Hiring Manager discussion on ranking reasons."
}
]
]

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3. Refined Requirements

```

{
"functional_requirements": [
"The system shall parse resumes and job descriptions to extract and match technical skills, work experience, and project details.",

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"The system shall calculate a candidate relevance score using a hierarchical weighting strategy where technical skills carry the highest weight, followed by relevant work experience, and then relevant projects.",
"The system shall evaluate project strength based on four explicit parameters: job description relevance, technical complexity, demonstrated impact, and evidence of practical implementation.",
"The system shall apply a scoring algorithm that allows high technical skill and project relevance scores to outrank higher years of experience with lower relevance.",
"The system shall generate a rationale report for every candidate score, detailing the primary factors and weighted attributes that contributed to their ranking.",
"The system shall support batch ingestion and asynchronous processing of large sets of candidate resumes.",
"The system shall include a data pipeline to clean, structure, and audit historical hiring data for historical bias prior to model training."
],
"non_functional_requirements": [
"Performance: The system shall complete processing and scoring of an individual resume within 3 seconds.",
"Accuracy: The system's automated ranking shall achieve at least 90% agreement with a human expert baseline on a representative validation dataset.",
"Fairness / Non-Discrimination: The system shall exclude protected attributes and proxies\u2014including gender, age, and college/university name\u2014from influencing candidate scores.",
"Constraint / Data Ethics: The system shall restrict company reputation attributes to contextual analysis only, preventing company names from directly inflating ranking scores.",
"Delivery Schedule: The Minimum Viable Product (MVP) shall be fully developed, validated, and ready for deployment within 8 weeks of project initiation."
]
}

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4. Requirement Quality Evaluation

```

{
  "after": {
    "clarity": 9,
    "completeness": 9,
    "overall": 9,
    "specificity": 9,
    "testability": 9,
    "unambiguity": 9
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  "before": {
    "clarity": 5,
    "completeness": 6,
    "overall": 5.4,
    "specificity": 5,
    "testability": 6,
    "unambiguity": 5
  },
  "explanation": "Requirements generated without clarification relied heavily on vague operational goals ('earn trust of HR', 'process quickly', 'solid projects') and deferred clarity to separate measurable criteria fields. Requirements generated with clarification explicitly defined scoring hierarchies, quantifiable thresholds (90% agreement, 3-second processing limit), strict bias-mitigation rules (prohibiting direct score inflation by college or company reputation), concrete project evaluation dimensions, and batch architecture needs. This significantly reduced ambiguity and enhanced testability and completeness.",
  "improvement_percentage": 66.7
}

```